

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 2 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 items, 1 point each — 8 points)

Write the letter of the best definition on the line beside each term. **Two definitions will not be used.**

- | | | | |
|----------|--------------------|----------|-------------------------------|
| E | 1. Domain | I | 5. Zero of a function |
| A | 2. Range | J | 6. Absolute maximum |
| C | 3. Parent function | B | 7. Horizontal asymptote |
| H | 4. Translation | F | 8. Piecewise-defined function |

- (A) The set of all output values a function produces.
(B) A horizontal line that a graph approaches at one or both ends without ever reaching it.
(C) The simplest function in a family, before any shift, flip, stretch, or compression.
(D) A stretch or a compression — a change to the *shape* of a graph.
(E) The set of all input values a function is allowed to take.
(F) A function defined by different rules on different stretches of its domain.
(G) The point where a graph crosses the vertical axis.
(H) A slide that moves a graph up, down, left, or right without changing its shape.
(I) An input value for which the output is 0.
(J) The largest output value a function reaches over its *entire* domain.

Part B — Multiple Choice (6 items, 2 points each — 12 points)

Circle the best answer.

- The point $(3, 10)$ is on the graph of f . Which statement says the same thing?
(A) $f(10) = 3$
(B) $f(3) = 10$ ← correct
(C) $f(3) = 3$
(D) $f(10) = 10$
- The graph of $y = (x + 5)^2$ is the graph of $y = x^2$ moved
(A) right 5
(B) up 5
(C) left 5 ← correct
(D) down 5
- A yearbook's profit is \$0 at prices of \$0 and \$60, and rises to a high of \$1800 at a price of \$30. Over which **prices** is the profit increasing?
(A) from 0 to 1800
(B) from 0 to 60
(C) from 30 to 60

(D) from 0 to 30 ← correct

4. Which statement about the graph of $y = 2^x$ is true?

(A) It crosses the x -axis at $x = 0$.

(B) It has no x -intercept: the curve approaches the line $y = 0$ without reaching it. ← correct

(C) It has an x -intercept somewhere far to the left.

(D) It has neither an x -intercept nor a y -intercept.

5. A garden center pays \$12 an hour for the first 40 hours of a week and \$18 an hour after that. The graph of a week's pay $P(h)$ has a **corner** at $h = 40$, where the rule changes. Which statement is true?

(A) The graph has a break at $h = 40$, so P is discontinuous there.

(B) $P(40)$ has two different values, one from each rule.

(C) P is continuous at $h = 40$: both rules give the same pay there, so the graph can be drawn without lifting your pencil.

← correct

(D) $h = 40$ is not in the domain of P .

6. A set of data curves **upward** as x increases. What does that observation tell you about which family fits?

(A) The data must be exponential.

(B) The data must be quadratic.

(C) Nothing at all.

(D) It rules out linear — and nothing more. You still have to check the differences and the ratios. ← correct

Part C — Short Answer & Computation (10 items, 5 points each — 50 points)

Show all work. An answer with no supporting work earns no credit.

1. A pool is being drained. After t hours it holds $W(t) = 3000 - 250t$ gallons, for $0 \leq t \leq 12$. (a) Find $W(4)$ and say what it means. (b) Solve $W(t) = 750$ and say what it means. (c) Write the domain and the range in interval notation.

(a) $W(4) = 3000 - 250(4) = 2000$ gallons after 4 hours

(b) $3000 - 250t = 750 \Rightarrow 250t = 2250 \Rightarrow t = 9$ hours

(c) domain $[0, 12]$, range $[0, 3000]$ ($W(12) = 0$, $W(0) = 3000$)

2. For each equation, describe the transformation from its parent function and give the range. (a) $y = (x - 4)^2 + 1$ (b) $y = 2^x - 5$ (c) $y = -x^2$

(a) x^2 shifted right 4 and up 1; range $[1, \infty)$

(b) 2^x shifted down 5; range $(-5, \infty)$

(c) x^2 reflected over the x -axis; range $(-\infty, 0]$

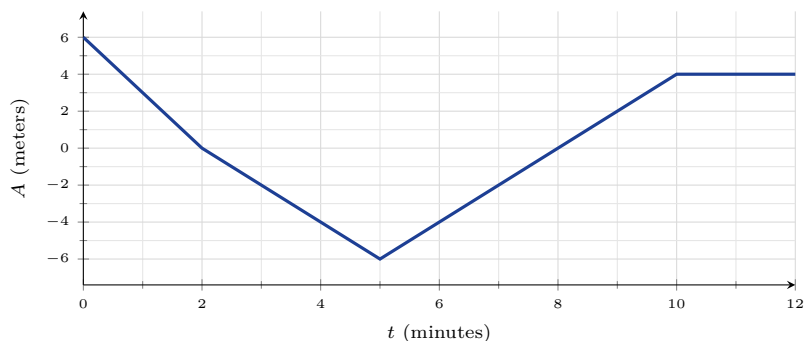
3. A parabola has vertex $(2, -3)$ and passes through the point $(4, 5)$. (a) Write its equation in the form $y = a(x - h)^2 + k$. (b) Verify your equation by substituting the point $(1, -1)$.

(a) $5 = a(4 - 2)^2 - 3$

$8 = 4a \Rightarrow a = 2$, $y = 2(x - 2)^2 - 3$

(b) $2(1 - 2)^2 - 3 = 2(1) - 3 = -1$ ✓

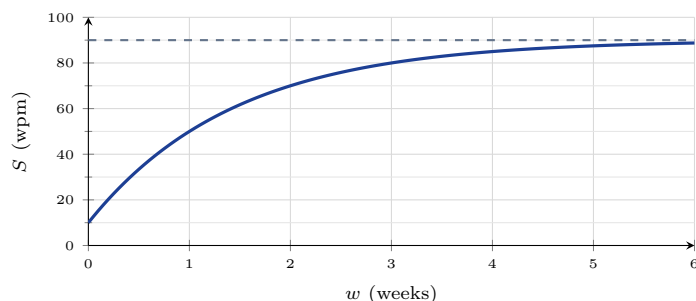
4. A drone flies out over a canyon. $A(t)$ is its altitude in meters **relative to the launch platform** ($A < 0$ means below the platform), t minutes into the flight.



- (a) Give the A -intercept and say what it means. (b) Give both zeros and say what a zero means here. (c) Give the intervals on which A is increasing, decreasing, and constant. (d) Give the absolute maximum and the absolute minimum — each as a **value and a location**.

- (a) **$(0, 6)$: the drone starts 6 m above the platform**
 (b) **$t = 2$ and $t = 8$: the drone is level with the platform**
 (c) **decreasing $(0, 5)$, increasing $(5, 10)$, constant $(10, 12)$**
 (d) **maximum 6 m at $t = 0$; minimum -6 m at $t = 5$**

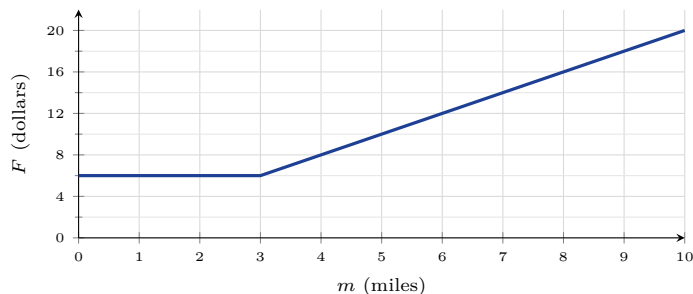
5. After w weeks of practice a typist's speed is $S(w) = 90 - 80(0.5)^w$ words per minute.



- (a) Write the equation of the horizontal asymptote. (b) Describe the end behavior as $w \rightarrow \infty$. (c) Will this typist ever reach exactly 90 wpm? (d) Could this typist ever reach 95 wpm? **Give a different reason for (c) and for (d).**

- (a) **$y = 90$**
 (b) **as $w \rightarrow \infty$, S levels off toward 90**
 (c) **No — the speed gets closer and closer to 90 but never arrives.**
 (d) **No — 95 is on the far side of the asymptote entirely.**

6. A rideshare charges a flat \$6 for the first 3 miles, then \$2 for each mile after that. The fare for a ride of m miles is $F(m)$.



(a) Find $F(2)$ and $F(7)$. (b) Name the boundary point. (c) Is F continuous or discontinuous at the boundary? **Justify from the graph.** (d) In one sentence, say what the two rules mean to a rider.

(a) $F(2) = 6$ dollars; $F(7) = 6 + 2(7 - 3) = 14$ dollars

(b) $m = 3$

(c) **Continuous: both rules give \$6 at $m = 3$, so the pieces meet.**

It is a corner, not a break — no pencil comes off the paper.

(d) **The first 3 miles cost a flat \$6; every mile after that adds \$2.**

7. For each table the x -values step by the same amount. Name the family — **linear, quadratic, or exponential** — and justify with the differences or the ratio.

(a)

x	0	1	2	3
y	4	7	10	13

(b)

x	0	1	2	3
y	5	10	20	40

(c)

x	0	1	2	3
y	2	5	14	29

(a) **first differences 3, 3, 3 — constant $\Rightarrow$ linear**

(b) **ratios 2, 2, 2 — constant $\Rightarrow$ exponential**

(c) **first differences 3, 9, 15; second differences 6, 6 $\Rightarrow$ quadratic**

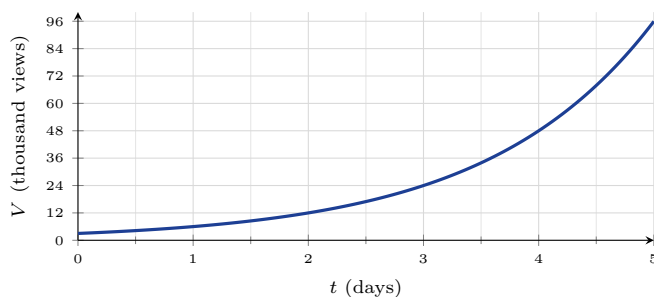
8. (a) Write $-2 \leq x < 7$ in interval notation, and say what each bracket or parenthesis means. (b) Write “all outputs greater than 3” in interval notation. (c) A field trip costs \$8 per student and the bus holds at most 30 students. Give the domain of $C(n) = 8n$ in this context, and explain why interval notation cannot express it.

(a) **$[-2, 7)$: the bracket includes -2 , the parenthesis excludes 7**

(b) **$(3, \infty)$ (∞ never takes a bracket)**

(c) **$\{0, 1, 2, \dots, 30\}$ — only *whole* students ride the bus, and an interval would include values like 17.4, which is not a domain value.**

9. A school video has $V(t) = 3 \cdot 2^t$ thousand views t days after it is posted.



(a) Find $V(5)$ **algebraically**. (b) Use the **graph** to find the day the video reaches 48 thousand views. (c) Name which method you used for each, and say why (a) is the easier one to trust here.

(a) $V(5) = 3 \cdot 2^5 = 3 \cdot 32 = 96$ thousand views

(b) **read across from 48 to the curve, then down: $t = 4$ days**

(c) **(a) is algebraic — substitution; (b) is graphical — a read.**

Substitution gives an exact value; a graph read is only as good as the grid.

10. Let $g(x) = 2^x - 8$. Give (a) the domain, (b) the y -intercept, (c) the zero, (d) the equation of the horizontal asymptote and the range, (e) the end behavior at both ends.

(a) all real numbers, $(-\infty, \infty)$

(b) $g(0) = 2^0 - 8 = 1 - 8 = -7$, so $(0, -7)$

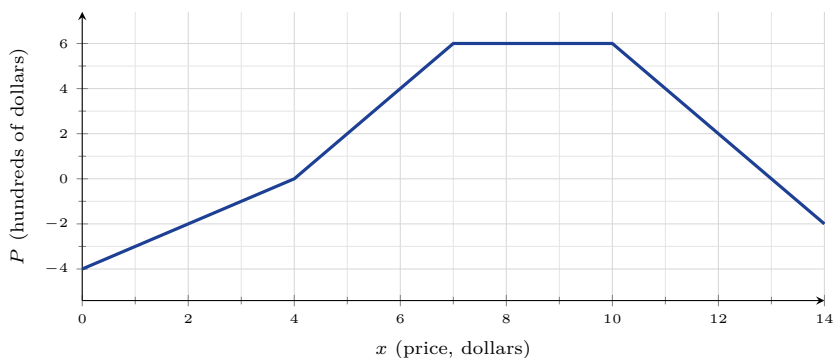
(c) $2^x - 8 = 0 \Rightarrow 2^x = 8 \Rightarrow x = 3$

(d) $y = -8$; range $(-8, \infty)$

(e) as $x \rightarrow -\infty$, g levels off toward -8 ; as $x \rightarrow \infty$, g rises without bound

Part D — Extended Response (2 items, 15 points each — 30 points)

1. **Read one graph completely.** A food truck's daily profit is $P(x)$, in *hundreds of dollars*, when its signature sandwich is priced at x dollars, for $0 \leq x \leq 14$.



- (a) Write the domain and the range in interval notation.

domain $[0, 14]$, range $[-4, 6]$

- (b) Give both zeros, and say what a zero means for this food truck.

$x = 4$ and $x = 13$

these are the two break-even prices — profit is \$0 there

- (c) Give the intervals on which P is increasing, decreasing, and constant.

increasing $(0, 7)$, constant $(7, 10)$, decreasing $(10, 14)$

- (d) Give the absolute maximum as a **value and a location**. Read the location carefully — it is not a single price.

maximum profit \$600 ($P = 6$), at every price from \$7 to \$10

the location is the *interval* $[7, 10]$, not one point

- (e) A consultant tells the owner: “charge as much as you can — more money per sandwich means more profit.” Use the graph to explain why that is wrong, and name the price you would recommend and why.

Wrong: past \$10 the graph falls, and past \$13 the profit is negative.
At \$14 the truck loses \$200 a day — too few people buy at that price.
Recommend any price from \$7 to \$10: each one earns the \$600 maximum.

2. **Choose a model, use it, and know where it stops.** A fundraiser's post is shared online. The table gives the number of shares on each of the first four days.

Day d	0	1	2	3
Shares S	8	24	72	216

- (a) Compute the first differences, the second differences, and the ratios. Name the family and justify your choice with the row that is constant.

first differences: 16, 48, 144 — not constant

second differences: 32, 96 — not constant

ratios: 3, 3, 3 — constant $\Rightarrow$ exponential

- (b) Write the model $S(d)$, then use it to predict the number of shares on day 5.

$$S(d) = 8 \cdot 3^d$$

$$S(5) = 8 \cdot 3^5 = 8 \cdot 243 = 1944 \text{ shares}$$

- (c) A student says: “the jumps keep getting bigger — 16, then 48, then 144 — so this is quadratic.” Every number the student wrote is correct. Explain what is still wrong with the conclusion.

Growing jumps rule out linear — and that is all they rule out.

Quadratic needs constant *second* differences; here they are 32 and 96.

The ratios are constant at 3, so the family is exponential.

- (d) This model predicts $S(15) = 8 \cdot 3^{15} = 114,791,256$ shares — more people than live in most countries. Does that mean the model is *wrong*? Answer in one or two sentences using the word **extrapolation**.

No — it fits the four days it was built from; day 15 is far outside them.

That is extrapolation: the model is read outside its window, not broken.